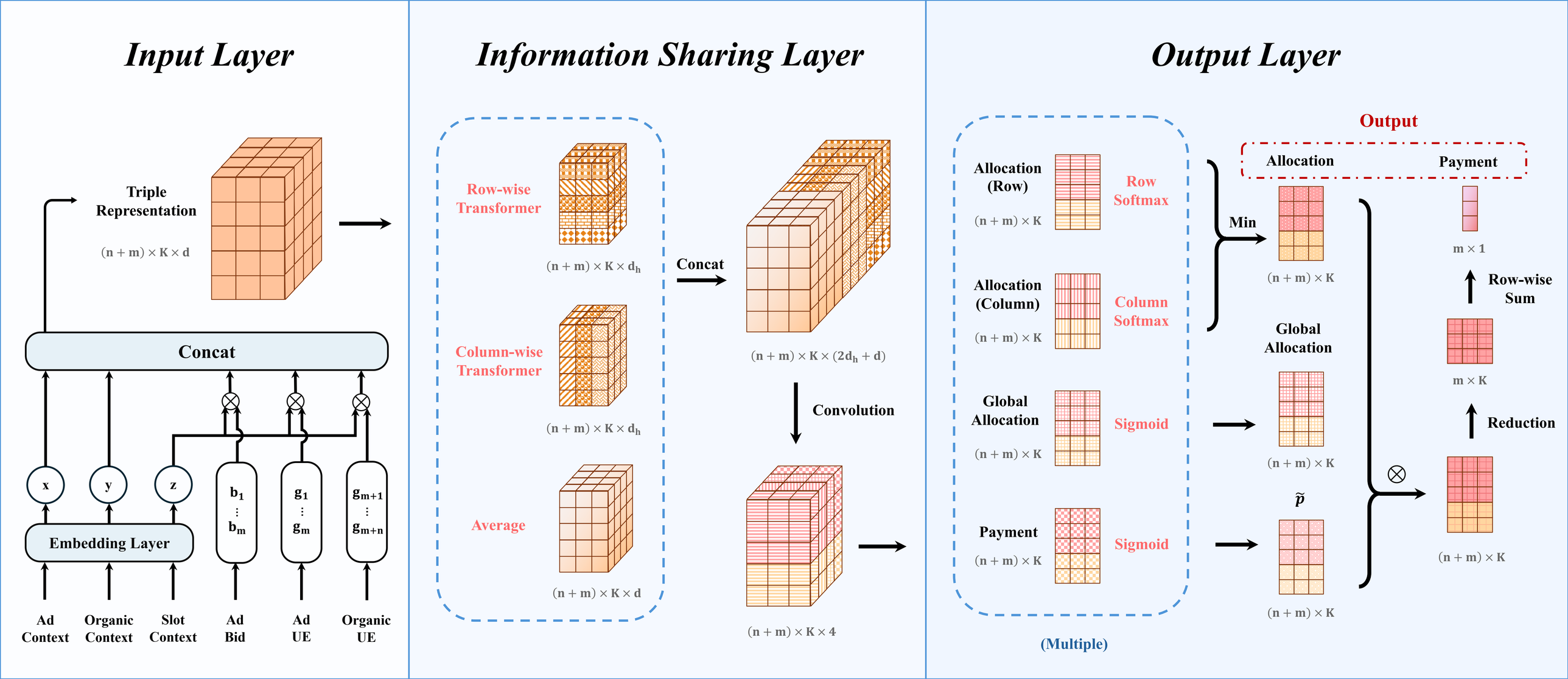
Input Layer
Triple Representation
(n + m) × K × d
Concat
x
y
z
b1 ⋮ bm
g1 ⋮ gm
gm+1 ⋮ gm+n
Embedding Layer
Ad Context
Organic Context
Slot Context
Ad Bid
Ad UE
Organic UE
Information Sharing Layer
Row-wise Transformer
(n + m) × K × dh
Column-wise Transformer
(n + m) × K × dh
Average
(n + m) × K × d
Concat
(n + m) × K × (2dh + d)
Convolution
(n + m) × K × 4
Output Layer
Allocation (Row)
(n + m) × K
Row Softmax
Allocation (Column)
(n + m) × K
Column Softmax
Global Allocation
(n + m) × K
Sigmoid
Payment
(n + m) × K
Sigmoid
(Multiple)
Output
Allocation
Payment
Min
(n + m) × K
Global Allocation
(n + m) × K
p̃
(n + m) × K
m × 1
Row-wise Sum
m × K
Reduction
(n + m) × K